9 Managing users with authentication

This chapter covers

- Implementing basic authentication
- Installing and configuring ASP.NET Core Identity
- Scaffolding and customizing the Identity UI

This and the next chapter look at managing users within a Razor Pages application. They look at two closely related topics: authentication and authorization. *Authentication* is the process of identifying who your user is. *Authorization* is part of the process of securing your application by limiting your user's access to only those parts of the application they are allowed to access.

This chapter concentrates on the nature of a *user* and how you can authenticate their identity, or ensure they are who they claim to be. Authentication is hard to get right. From a practical point of view, you need to provide mechanisms to capture and store a user's personal information, including something only they know, such as a password. Then you need to be able to *remember* the user on subsequent visits. Your application might also require you to use a secondary authentication mechanism, two-factor authentication (2FA), like a code sent by SMS. You might need to provide the user with features to reset their password if they have forgotten it or manage their profile information. What about verifying email addresses? Or managing account lockouts in the event of repeated incorrect attempts to log in? And you need to do all of this in a cryptographically secure manner.

To be truly secure, you need to keep up with ever-changing cryptographic standards and have a deep knowledge of potential attack vectors that allow bad actors (hackers) to hijack or clone a user's identity. For this reason, unless you are